

REPORT ON CHINOOK DATABASE

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Data Overview

The SQL queries developed in this project focused on analyzing customer activity, sales performance, product popularity, employee contribution, and revenue trends within the database.

I applied queries considering important SQL concepts such as:

- Joins
- Aggregate Functions
- Subqueries
- Common Table Expressions (CTEs)
- Window Functions
- Grouping and Filtering
- Revenue and Trend Analysis
- Query Optimization Techniques

Explanation of the 37 Queries

1. Customers with Invoices

Retrieves customers alongside their invoice records using an INNER JOIN.

2. Customers with Support Representative Names

Displays customers together with their assigned support representatives using LEFT JOIN.

3. Invoice Records with Customer Name and Date

Shows invoice details including customer names, invoice dates, and totals using INNER JOIN.

4. Artists and Their Albums

Lists all artists and their albums, including artists without albums using LEFT JOIN.

5. Employees Without Customers Assigned

Identifies employees who currently have no customers assigned to them using LEFT JOIN.

6. Total Sales per Country

Calculates total revenue generated from each country using JOIN, GROUP BY, ORDER BY.

7. Invoices Including Unmatched Records

Displays all invoices, including records without matching customers using LEFT JOIN.

8. Average Track Price

Calculates the average selling price of all tracks using AVG.

9. Tracks Priced Above Average

Finds tracks whose prices exceed the overall average track price using WHERE.

10. Customers Spending Above Average

Identifies customers whose total spending is above the average customer spending level using JOIN, WHERE.

11. Tracks Longer Than Average Length

Retrieves tracks with durations longer than the average track length using WHERE.

12. Employees Who Are Also Customers

Finds employees whose email addresses also exist in the customer table using EXISTS.

13. Customer Spending Rankings

Ranks customers based on their total spending using window functions using RANKS.

14. Employee Sales Rankings

Ranks employees according to the revenue generated from their assigned customers using ROW_NUMBER, RANK, PARTITION BY.

15. Invoice Rankings per Customer

Ranks invoices for each customer based on invoice total using ROW_NUMBER, RANK, PARTITION BY.

16. Top Customers per Country

Displays the top customers within each country based on total spending using ROW_NUMBER, RANK, PARTITION BY.

17. Top Products by Revenue

Identifies the tracks generating the highest revenue using GROUP BY, SUM.

18. Top Customers by Total Spending

Shows customers with the highest overall purchases using GROUP BY, SUM.

19. Top Genres by Revenue

Determines which music genres generate the most revenue using GROUP BY, SUM.

20. Top Tracks by Revenue

Lists tracks with the highest sales revenue using GROUP BY, SUM.

21. Monthly Revenue Trend

Analyzes revenue performance on a monthly basis using GROUPS BY DATE PARTS, LAG().

22. Yearly Revenue Trend

Shows revenue growth and performance by year using GROUPS BY DATE PARTS, LAG().

23. Month-over-Month Revenue Growth

Measures monthly revenue growth percentages using the LAG() function using GROUPS BY DATE PARTS, LAG().

24. Customer Purchase Frequency

Counts the number of purchases made by each customer using GROUP BY, SUM.

25. Revenue by Country

Calculates total revenue generated from each country using GROUP BY, SUM.

26. Top Artists by Revenue

Identifies artists contributing the highest revenue using GROUP BY, SUM.

27. Customers with Multiple Purchases

Finds customers who have made more than one purchase using GROUP BY, SUM.

28. Total Number of Albums per Artist

Counts the number of albums released by each artist using LEFT JOIN, GROUP BY, COUNT

29. Top Albums by Sales Volume

Displays albums with the highest number of units sold INNER JOIN, GROUP BY, SUM, TOP.

30. Most Frequently Purchased Tracks

Shows tracks purchased most frequently by customers using INNER JOIN, GROUP BY, COUNT, TOP.

31. Revenue Generated by Genre

Calculates revenue contribution by each genre using INNER JOIN, GROUP BY, SUM, TOP.

32. Top Employees by Sales Performance

Identifies employees generating the highest sales revenue using INNER JOIN, GROUP BY, SUM TOP.

33. Sales Generated by Media Type

Analyzes sales performance based on media type using INNER JOIN, GROUP BY, SUM.

34. Countries with the Highest Customer Count

Determines countries with the largest customer populations using group by, count, top.

35. States or Provinces with Highest Customer Count

Shows states or provinces containing the highest number of customers using GROUP BY, COUNT, TOP, WHERE IS NOT NULL.

36. Cities and States Generating Highest Revenue

Identifies cities and states producing the most revenue using INNER JOIN, GROUP BY, SUM.

37. Track Revenue Analysis

Calculates revenue generated by individual tracks and artists using INNER JOIN, GROUP BY, SUM.

Key Index Recommendations

1. CREATE INDEX IX_Invoice_CustomerId
ON Invoice(CustomerId);
2. CREATE INDEX IX_Invoice_InvoiceDate
ON Invoice(InvoiceDate);
3. CREATE INDEX IX_InvoiceLine_InvoiceId
ON InvoiceLine(InvoiceId);
4. CREATE INDEX IX_InvoiceLine_TrackId
ON InvoiceLine(TrackId);
5. CREATE INDEX IX_Customer_SupportRepId
ON Customer(SupportRepId);
6. CREATE INDEX IX_Album_ArtistId
ON Album(ArtistId);
7. CREATE INDEX IX_Track_AlbumId
ON Track(AlbumId);
8. CREATE INDEX IX_Track_GenreId
ON Track(GenreId);
9. CREATE INDEX IX_Track_MediaTypeId
ON Track(MediaTypeId);

Insight

The database analysis reveals strong relationships between customer purchasing behavior, employee performance, product popularity, and revenue growth.

Key findings include:

- Revenue is concentrated among a smaller group of high-value customers.
- Certain artists, tracks, and genres consistently outperform others.
- Revenue trends show opportunities for seasonal analysis and forecasting.
- Geographic analysis highlights countries and regions contributing the highest sales.
- Employee performance varies significantly based on customer sales activity.

Overall, the database structure effectively supports business intelligence reporting, sales monitoring, and customer behavior analysis.

Recommendations

- Introduce loyalty programs to retain high-value customers.
- Focus marketing on top-selling artists, genres, and tracks.
- Expand campaigns in countries and cities generating the most revenue.
- Improve employee performance through training and incentives.
- Promote high-performing products and review underperforming catalog items.
- Use monthly and yearly revenue trends for forecasting and planning.
- Implement indexes and optimize queries to improve database performance.
- Build dashboards and automate reporting for faster business insights.
- Improve data quality by validating customer and invoice records.